# Closed Composting Systems Will Likely Require High-Diversity Microbial Inoculants

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## Introduction

- Meilander et al. (2025) performed a mesophilic (20-45°C) composting experiment of human excrement (HE) over one year. The human excrement compost (HEC), is composed of excrement, toilet paper, and bulking material (i.e., pine shavings, straw, coconut coir).
- Microbes of unknown source (i.e., those that were not observed in the source HE and/ or the bulking material), accounted for the majority of the HEC microbiome composition.
- **We hypothesize that these unknown microbes originate from exposure to the external environment through air flow, insect activity, or other unknown modes of transfer.**

## Methods

- Individual buckets received bulking material and HE (Fig. 1A) as inputs. These inputs are classified as 'sources'.
- Features or taxa found in source inputs are classified as **intra-sourced** features (Fig. 1C).
- Features found in another bucket's source and are present in the current bucket are classified as **inter-sourced features**.
- Features not belonging to any source are labeled **extra-sourced**.

## Results

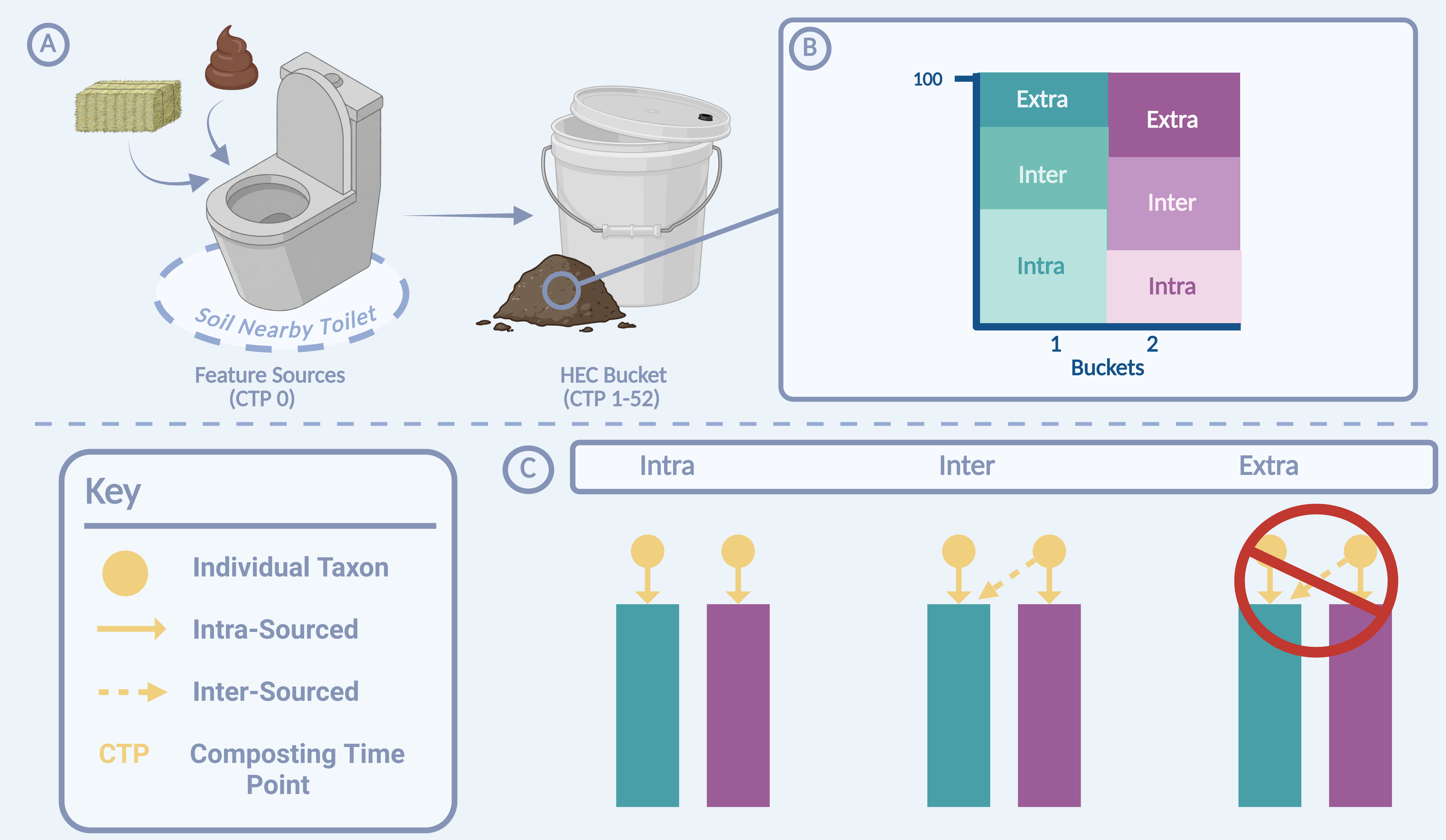


**Figure 1** : Human excrement composting process (A), example taxa bar-plot (B), and source classifications for externally-sourced microbes (C).

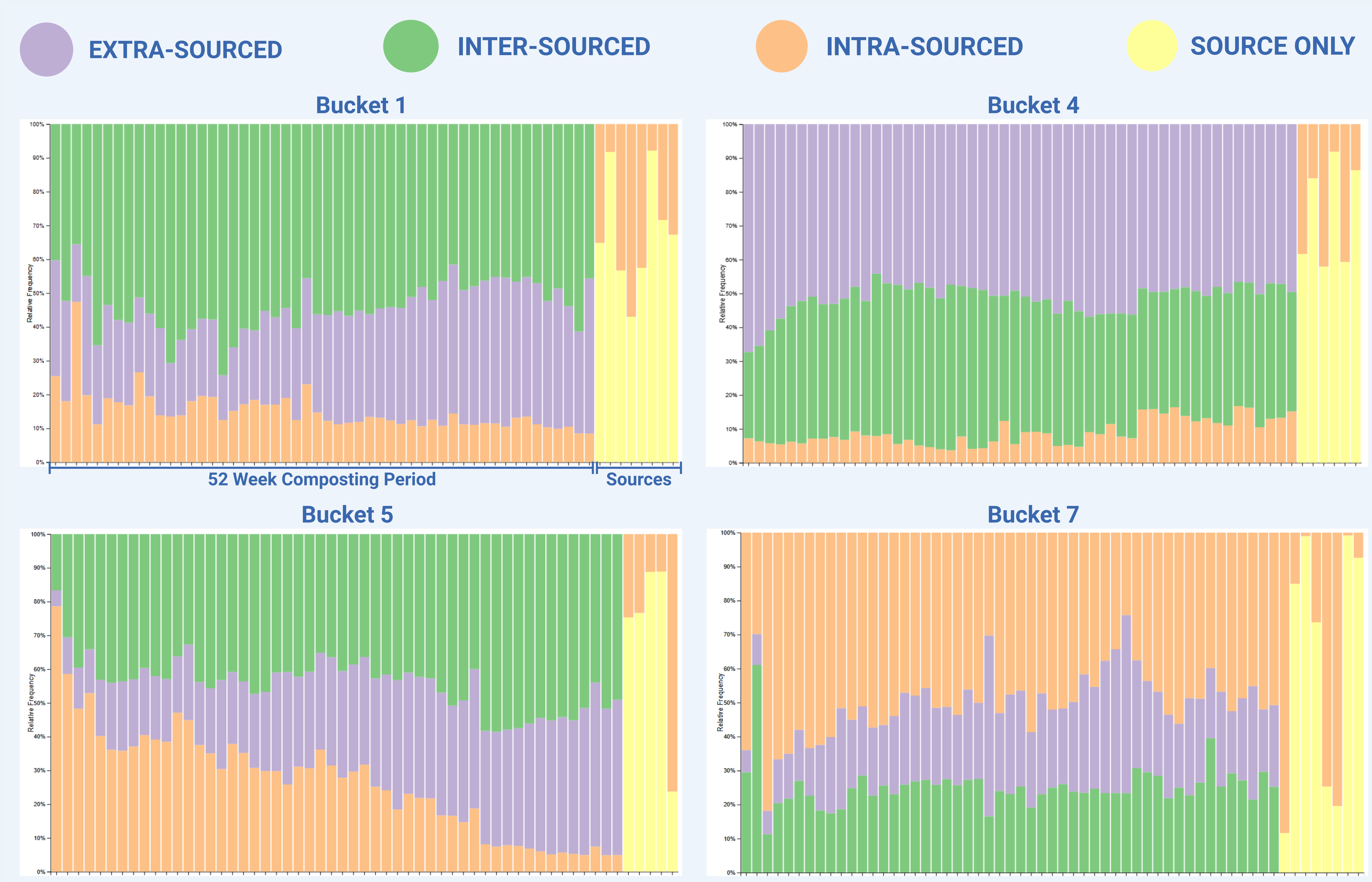


**Figure 2** : Per-bucket taxa bar plots show high percentages of externally sourced microbes over 52 weeks. Source only (yellow) features are represented at the tail end of the figures.

## Discussion

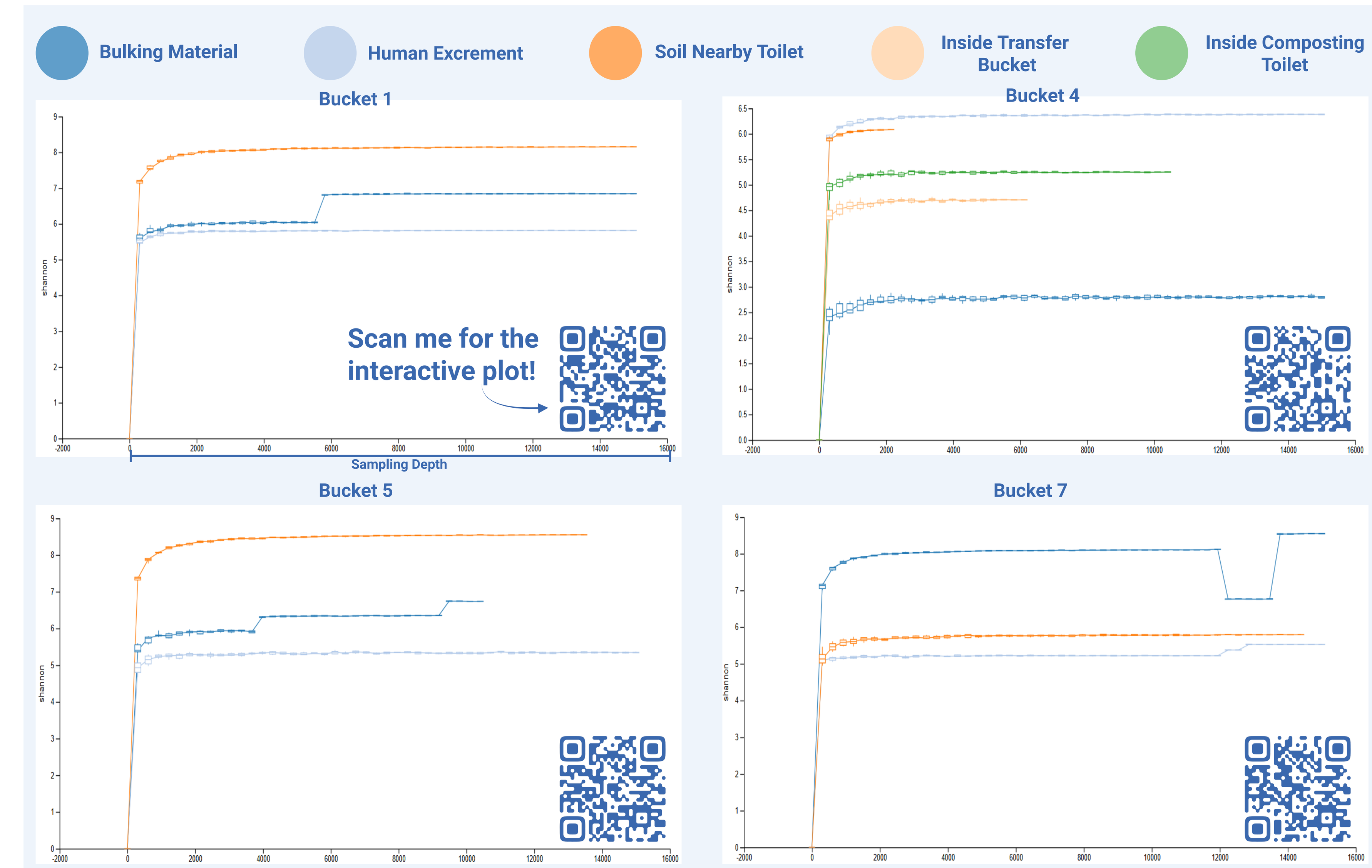


**Figure 3:** Alpha rarefaction plots show stable sequencing at our chosen sampling depth (15,000).

- We have **confirmed** these extra- and inter-sourced microbes have originated through external sources.
- Further analysis will determine the respective **function and origin** of individual microbes.
- Additionally, our analysis will aid in **determining the role** these unknown microbes perform in the composting process.

## Next Steps

- **Percent averages** of intra, inter, and extra sourced features over the 52-week composting period.
- Develop a **"parts list"** of microbes for future projects, particularly in closed-composting systems.

## Acknowledgments

The authors acknowledge the Interns-To-Scholars program at Northern Arizona University for funding this project, and State of Arizona Technology and Research Initiative Fund (TRIF) for funding the original experiment.

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